

Jimmy Gu

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PROFESSIONAL SUMMARY

Physics undergraduate with hands-on experience in computational research, data analysis, and scientific modeling. Background spans machine learning applications in biophysics research, data-driven consulting, and operational leadership at the United States Air Force Academy. Proven ability to work across technical and interpersonal domains in fast-paced, high-responsibility environments.

EDUCATION

Washington University in St. Louis

St. Louis, MO

Bachelor of Arts, Physics with Specialization in Biophysics; Biology Minor. Dean's List. GPA: 3.83.

December 2026

Relevant Coursework: Physics of Living Systems, Biophysics Laboratory, Quantum Mechanics for Engineers, Physics of the Brain, Electricity and Magnetism, Statistical Mechanics and Thermodynamics, Mechanics, Principles of Biology I and II, Organic Chemistry I and II, Introduction to Computer Science, Introduction to Computer Engineering

United States Air Force Academy

Colorado Springs, CO

B.S. (transferred). Superintendent's List: Top 10% physical performance, top 1/3 military standing. GPA: 3.90+.

June 2024

EXPERIENCE

Undergraduate Researcher

January 2025 – August 2025

Neuroaesthetics Research, Washington University

St. Louis, MO

- Investigated computational models of aesthetic perception in mosaic art, constructing a dataset of 982 images with multilevel statistical and segmentation features.
- Applied LASSO regression, entropy measures, and spectral analysis across 1,000+ image datasets to identify predictors of aesthetic ratings.
- Evaluated segmentation algorithms (SAM2, kernel graph cuts) for tesserae- and 60+ semantic-level feature extraction.
- Produced an abstract presented at the Spring 2026 Undergraduate Research Symposium, spanning 4 interdisciplinary topics bridging machine learning, human perception, and computational modeling.

Cadet

June 2023 – June 2024

United States Air Force Academy

Colorado Springs, CO

- Directed and coordinated mission planning for 100+ cadets while completing a full-time undergraduate curriculum.
- Designed and implemented simulation-based training programs for 100+ cadets, improving readiness, efficiency, and performance outcomes.

Consultant

January 2024 – August 2026

Consult Your Community (CYC), WashU Chapter

St. Louis, MO

- Analyzed fundraising data for 1,000+ donors at a local crime victim advocacy nonprofit, identified donor trends, and optimized outreach strategy.
- Served as social media consultant for a local interior design business, supporting 2 campaigns that increased engagement and visibility.
- Collaborated with 6 student consulting teams to deliver tailored recommendations across data analysis, client relations, and project management.

Founder and Board Member

December 2012 – June 2023

Journey of Hope

St. Louis, MO

- Founded and managed operations of a nonprofit organization for 10+ years, raising \$15,000+ in donations.
- Coordinated up to 20 volunteers and established strategic partnerships to sustain long-term organizational impact.
- Aligned resources, planning, and execution across 4 strategic initiatives using systems-level operational thinking.

SKILLS

Programming: Python, MATLAB

Methods and Analysis: Machine learning, Statistical modeling, Data regression, Spectral analysis, Systems engineering, Technical troubleshooting

Professional: Leadership, Project management, Technical writing, Presentations

ACTIVITIES AND INTERESTS

- Distance running: ~50 miles per week (Strava: [strava.com/athletes/59949384](https://www.strava.com/athletes/59949384))
- Judo and Brazilian Jiu-Jitsu